

HTTP is essentially a format for packaging data, and only hardcore web developers need to know more than the four parts of a request packet. The second entry field in that format is the header, containing the request method, which deserves some attention. The first version of HTTP had only one method, GET, which is sent by the client to request some resource (like a webpage) from the server. Some progress later, HTTP 1.1 has eight methods, and the second most popular request method is the POST method which is how elements like forms return user data to a web page. There are also the PUT and DELETE methods, which can be used for telling a database engine in the backend what to do (for example appending details in the database for creating a new user account).

## File Transfer Protocol (FTP)

The File Transfer Protocol was first developed by Abhay Bhushan in 1971. It is the predecessor of [HTTP](#), which was originally developed as an enhancement of [FTP](#), optimised for transfer of information on the scale of websites. FTP is still in use today for transferring web content, many hosting vendors' interfaces for uploading your website's files are driven by FTP transfers. There are many dedicated client softwares for FTP data transfer with a server, such as the open source FileZilla. Some WYSIWYG web page building software also include direct FTP upload functionality.

However, FTP was not designed to be secure, and has many vulnerabilities and holes as data is not encrypted. In fact, all transmissions are cleartext, so mere packet sniffing can easily land the username and password into malicious hands. To overcome this, FTP over Secure Shell (SSH) and FTP secured by SSL ([FTPS](#)) were developed.

## SMTP

SMTP stands for Simple Mail Transfer Protocol, which defines the how electronic mail is sent and received amongst the various mail servers on the networks of the internet. Mail servers are gateways between the client computer/application and the internet, that specialize in sending and receiving email. You, as a user, use an email client software (ex - Microsoft Outlook, Apple Mail, Google Inbox) to read and write your emails. The client software sends emails to the mail server using SMTP, however, it retrieves emails from the mail server using POP3 (Post Office Protocol version 3) or IMAP4 (Internet Message Access Protocol version 4). When you purchase the domain and hosting space for your website, you may also